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# SPECTRAL CANVAS

CSE 220

**REVOLVING  
IDEA**



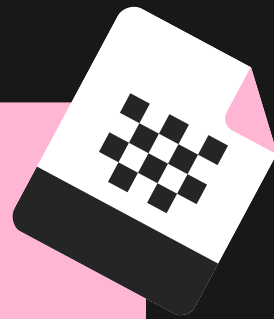
**SEND IMAGES AND  
TEXT OVER PHONE  
CALLS?!**

S  
I  
G  
N  
A  
L  
S

# THE PROBLEM



You want to send your friend an important document or image, but...



**THE INTERNET IS OUT.**

Physically printing it and sending is possible, but it would take hours and hours.

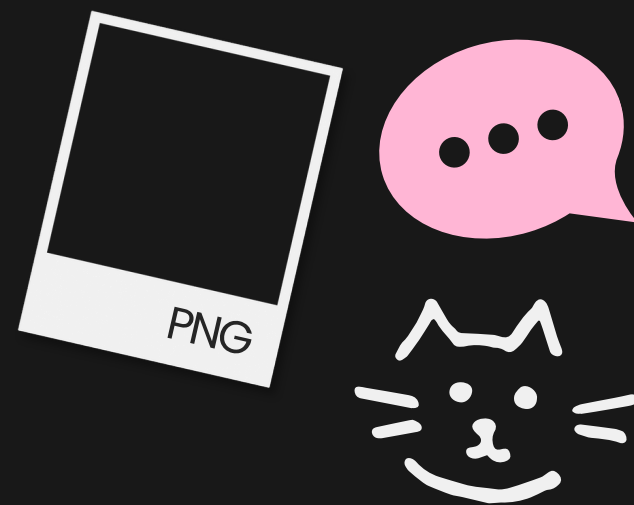


**WHAT TO DO NOW?**

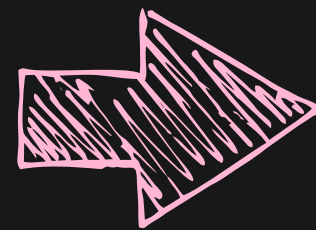
P.S. This idea was inspired from July 2024's situation, where internet often became unavailable even in times it was urgently required for posting notices and relief texts.

# OUR SOLUTION

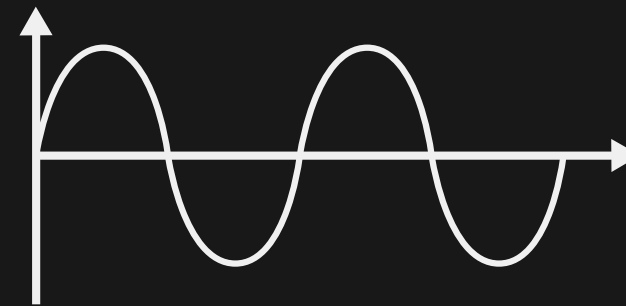
VER. A



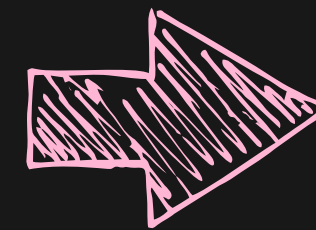
Images, Doodles  
or Text



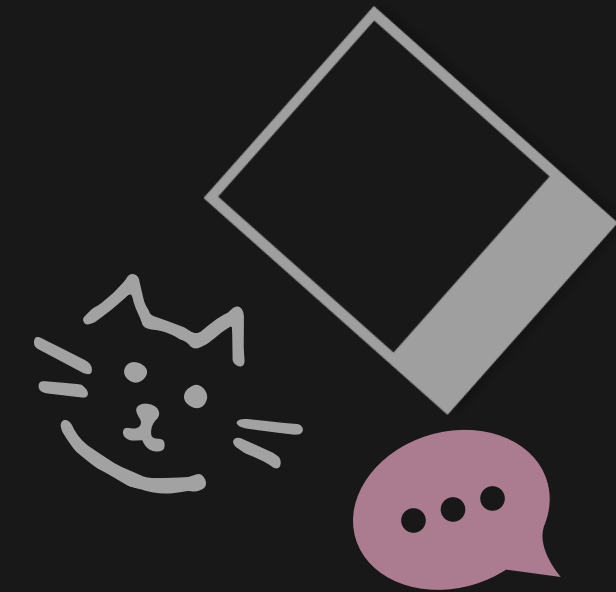
Encoder



Each row = frequency  
Each col = time frame



Decoder



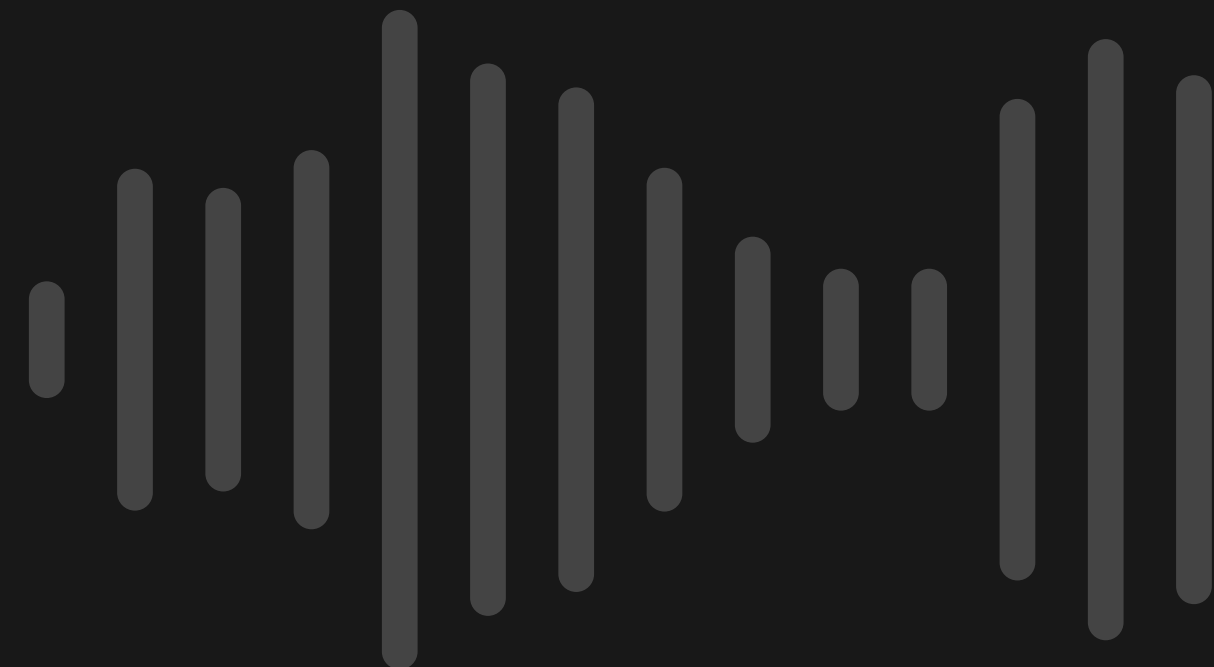
Recovered from  
phone call



We can send these things over a **PHONE CALL?** That's crazy!



# ISSUES WITH VER.A



Collapses over a **real GSM** call!



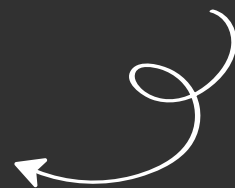
How?

We encoded every pixel in  
amplitude



**GSM 06.10** models 20 ms  
frames using spectral envelope

Amplitudes lost,  
frequencies preserved



Recovery drops  
to **75%**



# FIX: VERSION B!



Move the pixel out of amplitude, and into frequency

01

Choose **one tone**  
from 16 carriers  
between 700 and  
3200 Hz

4 bits/symbol

02

**Send** it over the  
GSM call

Enable GSM codec  
in Linphone

03

Demodulator  
takes **argmax**  
over tone bins

+ Hamming(7,4)  
interleaving and  
resync markers

04

**RESULT**

~**99%** exact  
recovery over  
call



# ALSO TEXT...

Bypasses the image pipeline entirely — separate codec

Each **UTF-8 byte**  
split into **two** 4-bit  
symbols (nibbles)

Each symbol maps  
to **1 of 16 tones**,  
spaced 2–5 kHz

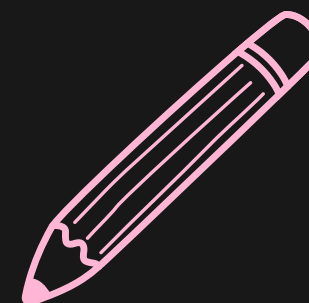
Every symbol is a **50  
ms Hann-windowed**  
sine burst

Decoder takes an **FFT  
peak per symbol**  
window, snaps to  
nearest tone

No image, **no lock/PIN**  
— just a direct byte-  
to-tone stream

Capped at **~1426 UTF-  
8 bytes** to keep the  
WAV under upload  
size limits

# THE WORKFLOW



## 1 UPLOAD

Upload an image, doodle, or write some text.

## 2 PARAMETERS

Grid Size, Gray tones / 16, B&W / color

## 3 ENCODE

Additionally, encrypt using caller + receiver number & PIN

## 4 PLACE CALL

Receive from Linphone app, mute and record the tones.

## 5 RECONSTRUCT

If encrypted, use same PIN to recover the image.

## 6 ENHANCE

(OPTIONAL) Use the model to enhance the quality

The demonstration will help us visualize this better!

# AND EVEN MORE...



 We built an experiments page where you can apply **different effects** to any image, and then measure how close the **LTI** reconstructs it! 

Low Pass

High Pass

Noise

Echo

Bandstop

Clipping

Butter-filter

Resampling



# LASTLY... MODELS!



Two networks: one restores, the other upscales!

**01**

## Restoration model

Resolves  
clipping,  
dead rows, and  
aliasing

**02**

## Upscaling model

Resolves  
quantisation and  
downscaling  
**INFERENCE!**

## DATASET



900 images in  
2K resolution

760 + 40  
Train-test

100  
validation



**THANK  
YOU!**

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**HOPE YOU FOUND IT**

**INFORMATIVE**

**&**

**ENJOYABLE!**